

Algorithm for file updates in Python

Project description

At my organization, access to restricted content is controlled with an allow list of IP addresses. The "allow_list.txt" file identifies these IP addresses. A separate remove list identifies IP addresses that should no longer have access to this content. I created an algorithm to automate updating the "allow_list.txt" file and remove these IP addresses that should no longer have access.

Open the file that contains the allow list

```
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:
```

Read the file contents

```
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()
```

Convert the string into a list

```
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

```
for element in remove_list:
```

Remove IP addresses that are on the remove list

```
for element in remove_list:

    # Create conditional statement to evaluate if `element` is in `ip_addresses`

    if element in ip_addresses:

        # use the `.remove()` method to remove
        # elements from `ip_addresses`

        ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

```
# Build `with` statement to rewrite the original file

with open(import_file, "w") as file:

    # Rewrite the file, replacing its contents with `ip_addresses`

    file.write(ip_addresses)
```

Summary

I created an algorithm that removes IP addresses identified in a `remove_list` variable from the "allow_list.txt" file. This algorithm involved opening the file, converting it to a string to be read, and then converting this string to a list stored in the variable `ip_addresses`. Then iterated through the IP addresses in `remove_list`. With each iteration, evaluated if the element was part of the `ip_addresses` list. If it was, applied the `.remove()` method to it to remove the element from `ip_addresses`. After this, I used the `.join()` method to convert the `ip_addresses` back into a string so that I could write over the contents of the "allow_list.txt" file with the revised list of IP addresses.